

Homework 1

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Exercise 1

Part (a)

Let A be a 1×1 complex matrix. Clearly, A is upper triangular, so A can be factorized as QTQ^* , where $Q = I$ and $T = A$, and so A has a Schur factorization. Now suppose that all $n \times n$ complex matrices have Schur factorizations. Let A be a $(n+1) \times (n+1)$ complex matrix, and let v_1 be a normalized eigenvector of A (with corresponding eigenvalue λ_1). We can construct an orthonormalized basis $\{v_1, \dots, v_{n+1}\}$ of $\mathbb{C}^{n+1}$ (using Gram-Schmidt, for instance). We now define U such that:

$$U = [v_1 \quad \cdots \quad v_{n+1}] = [v_1 \quad V_2]$$

Clearly, U is unitary. We then see that:

$$\begin{aligned} UAU^* &= \begin{bmatrix} v_1^* \\ V_2^* \end{bmatrix} A [v_1 \quad V_2] \\ &= \begin{bmatrix} v_1^* \\ V_2^* \end{bmatrix} [\lambda_1 v_1 \quad AV_2] \\ &= \begin{bmatrix} \lambda_1 & v_1^* AV_2 \\ 0 & V_2^* AV_2 \end{bmatrix} \end{aligned}$$

Let $B = V_2^* AV_2$ and $u = v_1^* AV_2$; since B is a $n \times n$ complex matrix, it has a Schur decomposition. Let that decomposition be $B = Q'T'Q'^*$. It then follows that:

$$\begin{aligned} UAU^* &= \begin{bmatrix} \lambda_1 & u \\ 0 & B \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0^* \\ 0 & Q' \end{bmatrix} \begin{bmatrix} \lambda_1 & u \\ 0 & T' \end{bmatrix} \begin{bmatrix} 1 & 0^* \\ 0 & Q'^* \end{bmatrix} \\ A &= U^* \begin{bmatrix} 1 & 0^* \\ 0 & Q' \end{bmatrix} \begin{bmatrix} \lambda_1 & u \\ 0 & T' \end{bmatrix} \begin{bmatrix} 1 & 0^* \\ 0 & Q'^* \end{bmatrix} U \end{aligned}$$

We then see that A has the Schur factorization $A = QTQ^*$ where:

$$Q = U^* \begin{bmatrix} 1 & 0^* \\ 0 & Q' \end{bmatrix}$$

$$T = \begin{bmatrix} \lambda_1 & u \\ 0 & T' \end{bmatrix}$$

since the products of unitary matrices are unitary. By induction, all square complex matrices have Schur factorizations.

Part (b)

Let A be a Hermitian complex matrix with Schur factorization $A = QTQ^*$. Then $A^* = (QTQ^*)^* = QT^*Q$. For this to be true, $T = T^*$, which means that T is a diagonal matrix with real entries. Since the eigenvalues of A are the diagonal entries of T , A has real eigenvalues.

Now let A have eigenvalues λ_1, λ_2 where $\lambda_1 \neq \lambda_2$ with corresponding eigenvectors v_1, v_2 . Then:

$$\begin{aligned} \lambda_1(v_1^*v_2) &= (Av_1)^*v_2 \\ &= v_1^*A^*v_2 \\ &= v_1^*Av_2 \\ &= v_1^*\lambda_2v_2 = \lambda_2(v_1^*v_2) \end{aligned}$$

It follows that $(\lambda_1 - \lambda_2)(v_1^*v_2) = 0$; since $\lambda_1 \neq \lambda_2$, $v_1^*v_2 = 0$, and so the eigenvectors are orthogonal.

Exercise 2

Part (a)

Let $A \in \mathbb{R}^{m \times n}$, with SVD $A = U\Sigma V^T$. We then see that:

$$\begin{aligned} A^T A &= (U\Sigma V^T)^T (U\Sigma V^T) \\ &= V\Sigma^T U^T U\Sigma V^T \\ &= V\Sigma^T \Sigma V^T \\ &= V\Sigma^2 V^T \end{aligned}$$

Since Σ is a diagonal matrix, Σ^2 must also be a diagonal matrix, with the diagonal entries being the squares of the diagonal entries of Σ . It follows that $A^T A$ has the SVD $V\Sigma^2 V^T$. Similarly:

$$\begin{aligned} AA^T &= (U\Sigma V^T)(U\Sigma V^T)^T \\ &= U\Sigma V^T V\Sigma^T U^T \\ &= U\Sigma \Sigma^T U^T \\ &= U\Sigma^2 U^T \end{aligned}$$

We see that AA^T then has the SVD $U\Sigma^2 U^T$.

Part (b)

Computing the SVD using $A^T A$ is numerically unstable. If A has condition number κ , $A^T A$ will have condition number κ^2 , so for ill-conditioned matrices, eigenvalue algorithms will perform poorly.

Part (c)

Let B such that:

$$B = \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix}$$

Let v be a right singular vector of A and u be the corresponding left singular vector (and therefore right singular vector of A^T), with singular value σ . Then:

$$\begin{aligned} x &= \begin{bmatrix} v \\ u \end{bmatrix} \\ Bx &= \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ u \end{bmatrix} \\ &= \begin{bmatrix} A^T u \\ Av \end{bmatrix} \\ &= \begin{bmatrix} \sigma v \\ \sigma u \end{bmatrix} = \sigma x \end{aligned}$$

Similarly:

$$\begin{aligned} y &= \begin{bmatrix} v \\ -u \end{bmatrix} \\ By &= \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ -u \end{bmatrix} \\ &= \begin{bmatrix} -A^T u \\ Av \end{bmatrix} \\ &= \begin{bmatrix} -\sigma v \\ \sigma u \end{bmatrix} = -\sigma y \end{aligned}$$

If A has singular values $\sigma_1, \dots, \sigma_n$, B then has eigenvalues $\pm\sigma_1, \dots, \pm\sigma_n$. The eigenvectors of B are as shown above. Additionally, the condition numbers of A and B are the same.

Exercise 3

Part (a)

Let A be a real symmetric $n \times n$ matrix. For $p \neq q$, let $A_{pq}^{(k)}$ and $G(p, q, \theta)$ such that:

$$A_{pq}^{(k)} = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$$

$$G(p, q, \theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

Then:

$$\begin{aligned} A^{(k+1)} &= G^T A^{(k)} G \\ &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} a & b \\ b & d \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} a \cos \theta - b \sin \theta & a \sin \theta + b \cos \theta \\ b \cos \theta - d \sin \theta & b \sin \theta + d \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} a \cos^2 \theta - b \sin(2\theta) + d \sin^2 \theta & -(b \cos(2\theta) + \frac{a-d}{2} \sin(2\theta)) \\ b \cos(2\theta) + \frac{a-d}{2} \sin(2\theta) & a \sin^2 \theta + b \sin(2\theta) + d \cos^2 \theta \end{bmatrix} \end{aligned}$$

For the off-diagonal entries to be zero, it must be the case that:

$$\begin{aligned} b \cos(2\theta) + \frac{a-d}{2} \sin(2\theta) &= 0 \\ \frac{a-d}{2} \sin(2\theta) &= -b \cos(2\theta) \\ \tan(2\theta) &= \frac{2b}{d-a} \end{aligned}$$

Part (b)

Let λ be an eigenvalue of $A^{(k)}$ with eigenvector v . Then:

$$\begin{aligned} A^{(k+1)}(G^T v) &= G^T A^{(k)} G G^T v \\ &= G^T A^{(k)} v \\ &= \lambda(G^T v) \end{aligned}$$

We see that λ is an eigenvalue of $A^{(k+1)}$ with the eigenvector $G^T v$. It follows that all iterates $A^{(k)}$ have the same eigenvalues as A .

Part (c)

Let $\text{off}(A) = \sum_{i \neq j} a_{ij}^2$. Note that $\text{off}(A^{(k+1)}) - \text{off}(A^{(k)}) = \text{off}(A_{pq}^{(k+1)}) - \text{off}(A_{pq}^{(k)})$, since all other entries are unchanged, and the diagonal elements of A_{pq} are also the diagonal elements of A . Then, since $\text{off}(A_{pq}^{(k+1)}) = 0$ and $\text{off}(A_{pq}^{(k)}) = 2a_{pq}^2$, it follows that $\text{off}(A^{(k+1)}) = \text{off}(A^{(k)}) - 2a_{pq}^2$. It follows that $\text{off}(A^{(k)})$ is monotonically decreasing.

Part (d)

If p, q are selected such that $|a_{pq}| = \max_{i \neq j} |a_{ij}|$, it follows that:

$$\begin{aligned} a_{pq}^2 &= \max_{i \neq j} a_{ij}^2 = \frac{n(n-1)}{n(n-1)} \max_{i \neq j} a_{ij}^2 \\ &\geq \frac{1}{n(n-1)} \sum_{i \neq j} a_{ij}^2 \\ &\geq \frac{1}{n(n-1)} \text{off}(A^{(k)}) \end{aligned}$$

We then see that:

$$\begin{aligned} \text{off}(A^{(k+1)}) &= \text{off}(A^{(k)}) - 2a_{pq}^2 \\ &\leq \text{off}(A^{(k)}) - 2 \left(\frac{1}{n(n-1)} \text{off}(A^{(k)}) \right) \\ &\leq \text{off}(A^{(k)}) \left(1 - \frac{2}{n(n-1)} \right) \end{aligned}$$

Since $0 < 1 - \frac{2}{n(n-1)} < 1$, the Jacobi algorithm converges linearly.

Exercise 4

```
1 def householder(A):
2     m, n = A.shape
3     Q = np.eye(m)
4     R = A.copy()
5
6     for j in range(n):
7         x = R[j:, j]
8         e1 = np.zeros_like(x)
9         e1[0] = 1
10        v = x + np.sign(x[0]) * np.linalg.norm(x) * e1
11        v_norm = np.linalg.norm(v)
12        if v_norm > 1e-14:
13            v /= v_norm
14            R[j:, j:] -= 2* np.outer(v, np.dot(v, R[j:, j:]))
15            Q[j:, :] -= 2* np.outer(np.dot(Q[:, j:], v), v)
16    return Q[:, :n], R[:, n, :]
```